

Twitter Sentiment Analysis

Multi-Class Emotion Detection from Text

1. Introduction

This project builds a machine learning pipeline for multi-class emotion classification on Twitter data. Rather than simple positive/negative sentiment, the task is to classify tweets into six emotions: sadness, joy, love, anger, fear, and surprise. The dataset is the Emotion Dataset from Kaggle, containing 20,000 English tweets pre-split into training (16,000), validation (2,000), and test (2,000) sets. Each tweet is labelled with a single emotion. The project is implemented end-to-end in a Jupyter Notebook using Python with scikit-learn, NLTK, and standard visualisation libraries.

2. Dataset

The dataset has two columns: **text** (the raw tweet) and **label** (integer 0-5 mapping to an emotion). There are no null values across any split. The class distribution is imbalanced: joy (33.5%) and sadness (29.2%) dominate, while surprise accounts for only 3.6%. This imbalance is a key factor in model evaluation.

Label	Emotion	Training Samples	Percentage
0	Sadness	4,666	29.2%
1	Joy	5,362	33.5%
2	Love	1,304	8.2%
3	Anger	2,159	13.5%
4	Fear	1,937	12.1%
5	Surprise	572	3.6%

3. Exploratory Data Analysis

EDA covered three areas. First, label distribution was plotted across all three splits to confirm consistent proportions and identify underrepresented classes. Second, text length analysis using histograms and box plots showed most tweets fall within 15-30 words, with no major length differences across emotions. Third, top-15 unigram frequency charts per emotion (excluding stopwords) revealed distinct vocabulary clusters: words like 'happy' and 'good' dominate joy, while 'afraid' and 'scared' dominate fear.

4. Text Preprocessing

A multi-step preprocessing pipeline was applied to all three splits:

- Lowercased all text
- Removed URLs, @mentions, #hashtags, and non-alphabetic characters
- Removed English stopwords (NLTK)

- Applied WordNet lemmatisation
- Filtered tokens shorter than 3 characters

Post-preprocessing analysis confirmed meaningful word count reduction via histogram overlays and scatter plots. Bigram analysis extracted the top 15 two-word combinations per emotion, revealing discriminative patterns like 'feel happy' (joy) and 'feel scared' (fear). Word clouds and a text feature heatmap provided additional visual summaries of the cleaned data.

5. Feature Extraction

TF-IDF vectorisation was applied with a vocabulary cap of 10,000 features, using both unigrams and bigrams with a minimum document frequency of 2. The vectoriser was fitted on training data only, then applied to validation and test sets to prevent data leakage.

6. Model Training and Comparison

Six classifiers were trained on the same TF-IDF features and compared:

Model	Train Acc	Val Acc	Test Acc
Logistic Regression	0.9253	0.8850	0.8815
Linear SVM	0.9676	0.8910	0.8910
Multinomial NB	0.8579	0.8410	0.8280
Random Forest	0.9999	0.8570	0.8470
Decision Tree	0.9999	0.8175	0.8175
KNN (cosine, k=7)	0.8196	0.7735	0.7690

Linear SVM achieved the best test accuracy at 89.1%, with a small train-validation gap indicating good generalisation. Random Forest and Decision Tree both overfit heavily (near-perfect training accuracy but significantly lower test scores). KNN with cosine distance performed the weakest at 76.9%.

7. Results and Error Analysis

The Linear SVM model was selected as the best performer. Per-class evaluation showed that joy and sadness achieved the highest F1 scores due to having the most training samples, while surprise had the lowest F1 (0.70) and recall (0.67), reflecting its severe underrepresentation at only 3.6% of the data. Confusion matrix analysis revealed that most misclassifications occur between semantically similar emotions: sadness/anger and love/joy share overlapping vocabulary, making them harder for the model to distinguish.

8. Tools and Libraries

Python 3, Jupyter Notebook, pandas, NumPy, matplotlib, seaborn, NLTK, scikit-learn, wordcloud.